

Persuading others in different communication media: appeals to logic, authority and emotion

Appeals to
logic, authority
and emotion

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Abstract

Purpose – Scholars of persuasion have long made a distinction between appeals to logic, emotion and authority- logos, ethos and pathos- but ideas developed to account for live face-to-face conversation processes must also be tested in new media. We aimed to test the effectiveness of these three strategies in one-to-one chats through different communication media.

Design/methodology/approach – With a $3 \times 3 \times 2$ between-subject factorial design, we tested these three strategies in one-to-one chats (female–female or male–male pairs) through three communication media: face-to-face, Skype video or Skype text. The persuasion scenario was adapted from prior studies in which students were presented with the idea of requiring a comprehensive exam as part of their degree. The participants were all undergraduate students of a major university in USA.

Findings – Our results showed trivial differences between female–female and male–male conditions. The logos appeal worked best overall in persuading the participants to change their reported attitudes. Additionally, the explanations provided by the participants for their own opinions were most like the persuasion scripts in the logos condition compared to the other two appeal conditions. Separately, participants indicated some disapproval of the pathos appeal in the text-based chat condition, although this did not seem to make a difference in terms of actual attitude change.

Research limitations/implications – One major limitation of our study is that our subjects are college students and therefore are not representative of Internet users in general. Future research should test these three types of persuasion strategies on people of diverse backgrounds. For example, while logos seems to be most effective strategy in persuading college students (at least in our study), pathos or ethos may be more effective when one attempts to persuade people of different backgrounds.

Practical implications – Although it is enough for a statistical test, our sample size is still relatively small due to constraints on time, personnel and funding. We also recognize that it is challenging both conceptually and empirically to compare the effectiveness of three persuasion strategies separately.

Social implications – Our findings suggest it is helpful to use fact-checking tools to combat disinformation in cases where users may not have sufficient domain knowledge or may not realize the need to identify or examine the given information. Additionally, it may require more effort to negate the impact of the disinformation spread than correcting the information, as some users may not only believe false information but also may start to reason in ways similar to those presented in the disinformation messages.

Originality/value – Past studies on online persuasion have limitedly examined whether and how communication media and persuasion strategies interact in one-to-one persuasion sessions. Our experiment makes an attempt to close this gap by examining the persuasion process and outcome in three different communication media and with three different persuasion strategies.

Keywords Persuasion, Logos, Ethos, Pathos, Communication media, Experiment, Face-to-face, Video chat, Computer-mediated communication

Paper type Research paper

Introduction

Online interactions involve complex processes of persuasion and influence. Persuasion is defined as “human communication that is designed to influence others by modifying their beliefs, values, or attitudes” (Simons, 1976, p. 21). Various aspects of online arguments affect



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persuasion processes and outcomes, including language features (Tan *et al.*, 2016; Li and Xiao, 2021), applied persuasion strategies (Hidey *et al.*, 2017; Chen *et al.*, 2021; Van Laer *et al.*, 2019; Seo *et al.*, 2013) and implied moral values (Kumar and Xiao, 2022). Communication contexts, such as the entry order of an argument (Mensah *et al.*, 2019), its position in a threaded discussion (Xiao and Mensah, 2022) and whether it is perceived as coming from an interpersonal source (Wilson and Djamasbi, 2022), also play important roles.

According to Aristotle, communicators may use three types of persuasive appeals to make their message convincing: logos, pathos, and ethos (Kennedy, 2015). Logos covers the logical appeal, based on offering facts, evidence and numbers. Pathos aims to stir the feelings and emotions of the audience, while ethos is an appeal to the authority or credibility of the presenter. The logos-pathos-ethos framework has been used to analyze social media content in varied persuasion scenarios, such as branding (Panigyrakis *et al.*, 2020), booking an Airbnb listing (Han *et al.*, 2019) and running a presidential campaign (Samuel-Azran *et al.*, 2015), illustrating the presence of Aristotelian rhetoric in online persuasion contexts.

Despite a growing body of literature on online persuasion research, our understanding of how the communication medium affects the effectiveness of different persuasion strategies remains limited. Prior research indicates that the online environment influences persuasion strategies and power in communication (Guadagno and Cialdini, 2002). However, as we detail in the related work section, findings about the effect of the communication medium on an online message's persuasion power are mixed (Bradner and Mark, 2002; Guadagno *et al.*, 2013). In this research, we explored whether Aristotle's three persuasion appeals – logos, pathos and ethos – perform differently in face-to-face conversation, text-based Skype chat and video-based Skype chat in a one-to-one interpersonal communication context. Given prior research suggesting that the persuasion outcomes in same-gender communications can be affected by the communication medium (Guadagno and Cialdini, 2002), we focused on same-gender communication and compared male and female pairs in our study.

Our investigation was conducted through a laboratory experiment that measured persuasion outcomes in interactions under different conditions. Three key variables were considered: the type of persuasion act in the communicated message (logos, pathos and ethos), the communication medium (face-to-face, text-based Internet chat, video-based Internet chat) and the gender of the participant (male–male vs. female–female interaction pairs).

The remaining sections of the paper are organized as follows: a review of studies related to online persuasion, a description of our experiment, including the design, task and participants, and finally, the presentation of the experiment results with a discussion of their implications.

Related work

Various theories have been developed to elucidate the mechanisms of persuading people and being persuaded. McGuire (1964) inoculation theory, for instance, explores how one's beliefs and attitudes may be fortified when facing a weak attack and actively defending one's convictions. This theory has found applications in combating misinformation on social media (Roozenbeek *et al.*, 2022; Lim and Ki, 2007).

The elaboration likelihood model (ELM) of persuasion, developed by Cacioppo and Petty (1989), delineates how attitudes can undergo change through either central or peripheral routes. The central route involves a careful consideration of presented information, while the peripheral route focuses on contextual information. In the presence of robust counterarguments, attitudes are expected to change through the central route. However, if one is unwilling or unable to consider the presented information, attitude change may still occur through the peripheral route influenced by contextual factors such as mood. In line with

this, small group research suggests that individuals in a face-to-face group setting may alter their opinions either due to persuasive information from other group members (Burnstein and Vinokur, 1975) or because they conform to the group majority, aligning with its social norms (Noelle-Neumann, 1980). The ELM has been employed in numerous studies examining contextual factors in online persuasion scenarios (e.g., Teng *et al.*, 2014; Lopes *et al.*, 2021).

The persuasive power of narrative has been widely acknowledged in the literature (Fisher, 1984). Narrative transportation theory posits that as individuals read or listen to a narrative, they become mentally immersed in the world depicted in the story, engaging their imagination, attention and emotions. Consequently, they are likely to align with the storyteller's perspective on the narrated world (Green and Brock, 2000). This theory is employed to elucidate why online reviews detailing people's purchase and usage experiences can be persuasive to others (Van Laer *et al.*, 2019).

Cialdini (2007) identified six contextual factors, or peripheral cues, that influence the persuasiveness of an act. These six factors are presented as six principles of influence: reciprocity, commitment/consistency, social proof, authority, liking and scarcity. These principles are applied in online persuasion studies, such as examining social influences on Twitter users (Yoo *et al.*, 2014).

Specific persuasion theories and models tailored for computer-mediated persuasion have also been developed. Investigating the role of Internet algorithms in online persuasion process, Zarouali *et al.* (2022) defined algorithmic persuasion as "*any deliberate attempt by a persuader to influence the beliefs, attitudes and behaviors of people through online communication that is mediated by algorithms*" (Zarouali *et al.*, 2022, p. 1078). They proposed the algorithm persuasion framework (APF), which focuses on how algorithmic techniques and algorithmic objectives affect the persuasion process and outcomes. In her disinformation study, Xiao (2021) built upon the persuasion knowledge model (PKM) developed by Friestad and Wright (1994) and presented a modified PKM explaining persuader-persuadee dynamics in online persuasion, emphasizing the role of persuasion knowledge in online episodes and outcomes. This model suggests key future research directions, including tests of which persuasion tactics are more potent than others on social media, how users become aware of persuasion acts behind disinformation and how persuasion knowledge differs across various social media environments.

Turning attention to the argument itself, the persuasion literature maintains that four aspects of language use are related to its persuasion power (Murphy, 2001): argument structure, argument content, comprehensibility of the text and credibility of the text. It has been demonstrated that the logical structure of arguments strongly influences consumers' beliefs about product claims (Munch *et al.*, 1993), and the use of rhetorical questions can impact a comment's persuasiveness (Petty *et al.*, 1981). Argument content is shown to be profoundly effective in the persuasion process (Allen, 1991; Chambliss, 1995). Comprehensibility refers to how easily the reader can grasp the text's meaning and perceive its relevance (Alexander and Jetton, 1996). A well-written text allows readers to process it deeply, increasing the chances of understanding and persuasion (Dole and Sinatra, 1998). However, prior work also indicated that strategic ambiguity might enhance the chances of persuasion because vague comments can be harder to reject (Hosman, 2002). Credibility of the text may be reflected in the use of quantifiers (Yalch and Elmore-Yalch, 1984), but when the text source has low credibility, the effect of quantification is negative for persuasiveness (Hosman, 2002).

Numerous studies have also analyzed the arguments in online persuasion processes to understand the use of persuasion strategies, including Aristotle's three types of appeals (logos, pathos and ethos). For instance, Hidey *et al.* (2017) examined these appeals in a sample of discussions from a subreddit called "Change My View" (CMV). In CMV discussions, participants offer their comments with the main purpose of changing the view of the original

post's author. By annotating these discussion dialogs, the researchers discovered that only 3% of the comments with premise sentences contain ethos, indicating attempts to establish credibility by emphasizing personal experience/expertise and reputation.

Several studies showed consistent findings as [Hidey *et al.*'s \(2017\)](#). [Chen *et al.* \(2021\)](#) characterized the usage of persuasion strategies in 22,621 posts flagged as containing misinformation in Sina Weibo, the largest microblogging site in China. The authors also analyzed 5,391 regular Sina Weibo posts for comparison. Besides Aristotle's three persuasion appeals, the authors included persuasion strategies based on other theoretical constructs such as narrative, Cialdini's influence principles, etc. They found that in both datasets, ethos strategies were the least commonly used persuasion strategies. In the study by Öcal and her colleagues ([Öcal *et al.*, 2021](#)), they annotated over 300 original posts in CMV discussions to explore the occurrence of the claims, rationales and the types of supporting statements for reasoning. In their analysis, only 5% of original posts contain statements that referred to the authors' domain expertise to establish the credibility of their views. Interestingly, in their study of Israeli politicians' Aristotelian rhetoric on Facebook and its reception during the 2013 elections campaign, [Samuel-Azran *et al.* \(2015\)](#) found a much heavier focus on the use of ethos (i.e., supporting candidates' credibility) and less emphasis on the use of pathos (i.e., emotional appeals) in the presidential campaign in Israel's multiparty political system compared to that in a two-party system like the United States. The observed inconsistency across existing studies about the use of ethos in social media implies the influence of the broad communication context on the prevalence and effectiveness of deployed persuasion strategies.

While there is a growing body of literature on online persuasion, our review of the related work suggests a lack of sufficient understanding of which of the three persuasion modes – logos, ethos and pathos – is more effective than the others online, and whether and how the distinctions among online and in-person communication media matter.

Findings regarding the effect of communication medium on the persuasion power of a message are mixed. [Guadagno *et al.* \(2013\)](#) suggest that the way information is displayed impacts how it is perceived. In their experiment, participants viewed the same information using one of three different options: paper, paper with graphs and computer with graphs. The authors found that information presented through a computer was more persuasive, especially for people who were novices in the domain. [Bradner and Mark's \(2002\)](#) experiment showed different results. In their experiment, participants engaged in a desert survival task using two types of computer-mediated communication: video and instant messaging. A desert survival task is a widely used group development training tool in which participants discuss how to allocate resources in a desert survival situation ([Lafferty *et al.*, 1978](#)). The researchers found that the communication medium did not affect the persuasion outcome.

Gender is also identified as a moderating factor in online persuasion. [Guadagno and Cialdini \(2002\)](#) conducted two studies comparing participants' attitudes after listening to or reading a series of arguments from a same-gender communicator via either email or face-to-face interaction. Their results showed that women were more likely to be persuaded by other women in face-to-face interactions than via email, whereas men showed no difference.

Experiment

To test whether logos, pathos and ethos hold the same persuasion power in different communication media, we conducted an experiment involving one persuasion session in a laboratory setting. With a $3 \times 3 \times 2$ between-subject factorial design, our experiment considered three factors. The first was the communication medium, which had three values: face-to-face, text-based Internet chat and video-based Internet chat. The second factor was the type of persuasion act, i.e., logos, pathos or ethos. With the third factor, we considered the

role of gender in the communication by comparing the persuasion outcomes in womanwoman pairs with those in man-man pairs. The details of our experiment are described below.

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Persuasion session

The experiment was conducted between October 2018 and April 2019. Our participants were undergraduate students who were enrolled in intro-level Psychology courses at the time of the experiment and were given course credits by participating in the study. After giving informed consent, the participant asked to take on the role of an interviewer in a recorded session, then provided an interview topic and a list of interview questions. The participant was instructed to only use the listed interview questions when interviewing the other participant. (The “other participant” was a research assistant.)

Our interview setup was adapted from [Guadagno and Cialdini's \(2002\)](#) persuasion experiment. Specifically, each participant was told that this interview procedure was part of an effort to gather information about genuine student opinions on the idea of institutionalizing a comprehensive exam in Psychology major, and that we were asking some students (including the participant) to interview other students in order to avoid biasing their responses through the presence of a representative of the department. Under the cover story, then, the interviewee was not trying to persuade the participant specifically, just record their opinion – but we were primarily interested in persuasive effects on the participant that might arise as a result of incidental exposure to the interviewee's argument in favor of comprehensive exams.

We developed three scripted answers to the interview questions using logos, ethos or pathos, respectively. Contextualized in this task, the persuasion direction in the scripts was to persuade participants to be in favor of adding the comprehensive exam as a new degree requirement at the Psychology department. Hence, in our design, the logos appeal argued on practical grounds that a comprehensive exam would strengthen students' job applications. In attempt to arouse the recipient's positive emotion, the pathos appeal claimed that the actor would feel pride and satisfaction upon completing such a milestone. The ethos appeal claimed that the actor knew an expert academic advisor who believed that comprehensive exams are useful. Prior studies suggest that the length of an online message can affect its perceived persuasiveness ([Xiao and Khazaei, 2019](#); [Wang et al., 2020](#); [Deng et al., 2020](#)). We designed the three appeals to be of the same length in terms of the total number of words in a script. The full scripts of the three appeals are in [Appendix](#).

The wording of all three scripts was identical across media conditions. In the face-to-face condition, the actor delivered the lines from memory while sitting across a table from the participant. In the video condition, the actor delivered the lines via a videoconferencing platform. In the text condition, the actor entered the lines as plain text into the chat feature of the same videoconferencing platform, with no audio or video component.

The research assistants who played the role of interviewee in our study were undergraduate students majoring in Psychology. They went through a training process and a couple of pilot sessions to memorize the script content for each appeal and to act naturally when offering the interview responses. As a result of the training, they gave their responses with a consistent speed in each communication medium condition and paused at the same places when giving their responses.

At the end of the persuasion session, the subjects filled out an online questionnaire which probed their impression of the interviewee's opinion regarding the comprehensive exam idea, their own opinions and explanations and their impressions of the interviewee. After that, we debriefed to the subject about our deception, i.e., the comprehensive exam idea was fake. The subjects had the option to withdraw and have their data removed from the study without penalty. We also probed whether the subject felt that it was fake during the experiment.

Independent variables

Our experiment has three factors, hence, three independent variables. The first variable is the communication medium which has three values: face-to-face, text-based Skype chat and video-based Skype chat. The second variable is the type of persuasion mode, and it also has three values: logos, pathos and ethos. The third variable is the gender combination of an experiment session, and it has two values: male–male versus female–female pairs.

Dependent variables

We measured and compared the outcome of the persuasion session in different conditions. Adopting [Miller's \(2013\)](#) perspective on persuasion, a persuasion act is successful if it leads to one of three attitudinal outcomes for its targeted individual: the individual who does not have a pre-existing view regarding the issue conveyed in the comment establishes a view (view establishment), the individual's pre-existing view regarding the issue is reinforced (view reinforcement) and the individual changes his/her pre-existing view regarding the issue (view change). To measure the outcome of a persuasion session, we collected and compared the participant's opinion regarding the comprehensive exam topic before and after the persuasion session through a pretest and a post-test. The pretest was administered earlier in the semester as part of a larger online survey featuring eligibility requirements for several psychology studies simultaneously. This larger survey is a standard part of the subject pool experience. Participating in this pretest survey counts toward course credit for introductory-level psychology students, and most of them choose to do it because it is a straightforward way to increase the number of studies that appear to them in the online subject pool system. They remain unaware of which pretest questions made them eligible for which studies. The pretest question of this study is: "Suppose the Syracuse University psychology department was considering adding a comprehensive exam as a new degree requirement. This hypothetical policy would only apply to future students, not currently enrolled students. Please indicate whether you would be opposed to or in favor of a comprehensive exam for future students: Completely opposed, Opposed, Slightly opposed, Neutral, Slightly in favor, In favor, Completely in favor." All participants who responded to this item were eligible to participate, except for those who indicated being "in favor" or "completely in favor" (the two least common responses, representing less than 5% of the pool). The pretest was administered using an online survey. No other items from the pretest were relevant to the present research.

The post-test was taken right after the interview in the form of an online questionnaire which included the question, "What is your opinion regarding the interview topic – adding a comprehensive exam as a new degree requirement at the Psychology department? Please choose one option from below: Completely opposed, Opposed, Slightly opposed, Neutral, Slightly in favor, In favor, Completely in favor". The difference in the responses to these two items suggested the outcome of the persuasion session. Coding each option as a number (e.g., Completely opposed = 1 versus Completely in favor = 7). This outcome can be represented numerically in two ways – as the presence or absence of any reported attitude change from baseline toward the research assistant's advocated position (post-test – pretest >0) or as the magnitude of such reported attitude change (post-test – pretest). We analyzed the data both ways.

The order of the post-test questions was critical as the reading and answering of a question may affect the participant's awareness that the interviewee was trying to persuade him/her. So, we asked those questions about the participant's perception of the interviewee before probing their awareness to the experiment's purpose. In addition, the subject could not go back to change their responses to the questions once they moved to the next page of the questionnaire.

To situate the participant in the scripted persuasion context, our first question in the post-test was, “What is the interviewee’s opinion regarding the interview topic—adding a comprehensive exam as a new degree requirement at the Psychology department? Please choose one option from below: Completely opposed, Opposed, Slightly opposed, Neutral, Slightly in favor, In favor, Completely in favor” (Q1). Then on the next page, the subject was asked to offer their *own* opinion on this comprehensive exam (Q2; quoted previously) and then provide their reasons – “Please explain your opinion. What are the two or three most important reasons of your indicated opinion above?” (Q3). We examined whether the reasons provided by the subject were similar to those provided in the interview script by measuring the semantic similarity between the written response of this question and the script used in that session.

The post-test questionnaire also included questions to probe the other beliefs and attitudes related to the persuasion processes, such as the subject’s liking of the persuader and their awareness of the persuasion attempt. Then the remaining questions were for the subject’s demographic information such as gender, age, academic status, etc.

Result

The lab experiment was carried out in two semesters from Oct. 2018 to May 2019 at a major university in the New York state. In total, 78 male and 118 female undergraduate students from the Department of Psychology were recruited. No subject withdrew from the study after the debriefing, but some students said that they felt the comprehensive exam idea was fake when asked during the debriefing, so their data were excluded from subsequent analysis. In total, we have 165 participants in our data analysis (60 male and 105 female undergraduate students). (Here, we identify the participant gender distribution based on their responses to the corresponding question in the post-test questionnaire.)

Persuasion check

To examine whether our persuasive appeals, taken together, indeed played a persuasive role on the subject’s opinion, we measured the difference in a subject’s opinion regarding the comprehensive exam between their response in the pretest and their response in the post-test. The position of our designed persuasion appeals is to support this exam idea. Before the persuasion attempt, the participants’ mean opinion regarding the comprehensive exam requirement for the Psychology degree was between slightly opposed and neutral ($M = 3.29$, Standard Deviation (SD) = 1.16). Women and men did not significantly differ from one another (Female: $M = 3.38$, SD = 1.12; Male: $M = 3.12$, SD = 1.21), $p = 0.17$. The participants showed greater support for the comprehensive exam idea after the interview experiment in both groups (Female: $M = 5.26$, SD = 1.37; Male: $M = 5.02$, SD = 1.50). Overall, the persuasion appeals were successful when comparing the subjects’ post-persuasion opinion ($M = 5.18$, SD = 1.43) and the prepersuasion opinion ($M = 3.29$, SD = 1.16), $t(164) = 13.67$, $p < 0.001$.

The effectiveness of our distinct persuasion appeals

To test the effect of the manipulations on attitude change, we built regression models predicting the change of the participant’s opinion. The models considered the factors of communication medium, persuasion appeal and gender, and used the participant’s opinion in the pre-test as a covariate. We did not observe a significant three-way interaction between medium, appeal and gender, $F(4, 146) = 1.77$, $p = 0.14$. Furthermore, gender did not significantly interact with either medium or appeal individually, $F < 2.81$, $p > 0.063$, and men and women did not significantly differ from one another, $b = 0.17$, standard error (SE) = 0.24,

$p = 0.46$. Therefore, our further analysis did not differentiate the male–male and female–female conditions. The overview of the participants in different experimental conditions, ignoring gender, is presented in [Table 1](#).

We did not observe a significant interaction between medium and appeal, $F(4, 154) = 0.76$, $p = 0.55$. The three communication medium conditions did not significantly differ from one another, $b < 0.12$, $SE > 0.27$, $p > 0.67$. However, the logos condition resulted in significantly more attitude change than either the ethos condition, $b = 0.79$, $SE = 0.28$, $p = 0.005$ or the pathos condition, $b = 0.69$, $SE = 0.27$, $p = 0.010$. The ethos and pathos conditions did not differ significantly from each other, $b = 0.10$, $SE = 0.27$, $p = 0.71$.

We also tried treating attitude change as a binary variable, where any positive attitude change (post-test attitude $>$ pretest attitude) was contrasted with no positive attitude change. Most (132/165; 80%) participants exhibited some attitude change, so the “not persuaded” outcome was rare for this analysis, hindering its statistical power. The results appeared similar to the approach focusing on magnitude of change. The medium manipulation did not make a significant difference, odds ratio (OR) < 1.68 , $b < 0.52$, $SE > 0.49$, $p > 0.32$. The logos condition resulted in significantly more attitude change than the ethos condition OR = 4.39, $b = 1.48$, $SE = 0.57$, $p = 0.010$. Although the logos condition appeared to be the most persuasive overall, it was not significantly different from the pathos condition in this analysis, OR = 2.44, $b = 0.89$, $SE = 0.57$, $p = 0.12$. The ethos and pathos conditions did not significantly differ, OR = 1.80, $b = 0.59$, $SE = 0.45$, $p = 0.19$.

Participant’s reasoning of their opinions

To examine whether and how the persuasive appeal and communication medium affected the participants’ reasoning of their own opinions, we measured the semantic similarity between each participant’s written response explaining their opinion (Q3) and the interview script used in the experiment, and fit linear regression models predicting the similarity based on medium, appeal and gender. We first vectorized these texts using a Universal Sentence Encoder ([Cer et al., 2018](#)) on TensorFlow ([Abadi et al., 2016](#)) without any preprocessing steps, and then calculated the inner product of the two text vectors using the inner function in NumPy ([Walt et al., 2011](#)). This inner product is our index of semantic similarity between the two texts.

In terms of the interaction effect, our regression analysis showed that there was no significant three-way interaction between medium, appeal and gender, $F(4, 147) = 0.25$, $p = 0.91$. Gender did not significantly interact with either medium or appeal individually, $F < 0.10$, $p > 0.91$. There was also no significant interaction between medium and appeal, $F(4, 155) = 0.64$, $p = 0.63$. Furthermore, the three medium conditions did not significantly differ from one another, $b < 0.02$, $SE > 0.01$, $p > 0.50$. Men and women did not significantly differ from one another either, $b = 0.02$, $SE = 0.01$, $p = 0.13$. Interestingly, the logos condition resulted in significantly more similarity than either the ethos condition, $b = 0.15$, $SE = 0.02$, $p < 0.001$, or the pathos condition, $b = 0.14$, $SE = 0.02$, $p < 0.001$. The ethos and pathos conditions did not differ significantly from each other, $b = 0.01$, $SE = 0.02$, $p = 0.52$. This observation suggests that the logo appeal excels at changing the recipient’s attitude but also tuning their reasoning style to adopt more of the language used in the appeal.

Table 1.
The number of
participants in
different experiment
conditions

	Live	Text	Video	Total
Etho	18	24	12	54
Logo	17	14	21	52
Patho	20	16	23	59
Total	55	54	56	165

Source(s): Table by authors

However, there is an alternative explanation to this observation. Specifically, as subjects are college students it is possible that they tend to provide justifications in a logical way regardless of condition, resulting in higher measured similarity with the logo script. To examine the validity of this explanation, we measured the semantic similarity between each participant's response and the logos script regardless of the appeal condition the participant was in. Results for this version of semantic similarity were very similar. Specifically, there was no significant three-way interaction between medium, appeal and gender, $F(4, 147) = 0.78, p = 0.54$. Gender did not significantly interact with either medium or appeal individually, $F < 0.12, p > 0.88$. There was also no significant interaction between medium and appeal, $F(4, 155) = 0.57, p = 0.68$. With respect to the main effects, we did not observe a statistically significant difference among the three medium conditions, $b < 0.02, SE > 0.02, p > 0.53$. Men and women did not significantly differ from one another either, $b = 0.03, SE = 0.02, p = 0.16$. Interestingly, the logos condition resulted in significantly more similarity than either the ethos condition, $b = 0.07, SE = 0.02, p < 0.001$ or the pathos condition, $b = 0.04, SE = 0.02, p = 0.036$. The ethos and pathos conditions did not differ significantly from each other, $b = 0.03, SE = 0.02, p = 0.14$. This finding shows that the alternative explanation that participants ordinarily rely on language similar to the logos script regardless of condition does not fully account for the effect of condition on similarity. In other words, people in the logos condition seemed to use more similar language to the logos script than people in the other conditions did.

Participant's liking of the persuader

One of Cialdini's six influence principles (Cialdini, 2007) is Liking – if an individual likes the persuader, then the individual is more likely to be persuaded. We assessed participants' opinions on the following statements through 5-Likert response scales. "I feel that I would get along well with the person that I interviewed" (Q4). "I could become very good friends with the person that I interviewed" (Q5). "The person that I interviewed appears to be very likeable" (Q6). We measured each participant's liking of the interviewee by averaging the results of these three items (Q4, Q5 and Q6; Cronbach's alpha = 0.86). This measure was weakly and not significantly correlated with the participant's degree of attitude change from pretest, $r(163) = 0.14, p = 0.077$.

We then tested the effect of the manipulations on liking by fitting linear regression models predicting liking based on medium, appeal and gender. In terms of the interaction effect, we did not observe significant three-way interaction between medium, appeal and gender, $F(4, 147) = 2.95, p = 0.53$. Gender did not significantly interact with either medium or appeal individually, $F < 0.52, p > 0.59$. Also, there was not a significant interaction between medium and appeal, $F(4, 155) = 0.14, p = 0.97$. With respect to the main effects, the Live condition resulted in significantly more liking than the text condition, $b = 0.76, SE = 0.18, p < 0.001$. Similarly, the video condition resulted in significantly more liking than the text condition, $b = 0.76, SE = 0.19, p < 0.001$. The Live and Video conditions did not significantly differ, $b = 0.00, SE = 0.18, p = 1.00$. The appeal conditions did not significantly differ from one another, $b < 0.13, SE > 0.17, p > 0.50$. Also, women reported significantly more liking of the female persuaders than men did of the male persuaders, $b = 0.55, SE = 0.16, p < 0.001$.

The perceived persuasiveness of the persuader

Many previous studies predefined the strength of a persuasion attempt and used it as an experimental condition (i.e., strong vs weak arguments) (Griskevicius et al., 2010; Moons et al., 2009). In contrast, we measured the perceived strength of the persuasion attempt in a session and examined whether and how the manipulations affected this perception. Our measurement was based on the response of several questions in the questionnaire. Specifically, we asked the

participant to indicate their agreement (on a 5-point Likert scale) with the following statements. “The arguments that the interviewee presented were persuasive” (Q7). “The interviewee made a strong argument for comprehensive exams” (Q8). By averaging the results of these two questions, we measured the participant’s subjective judgment of how persuasive the appeal was ($r = 0.78$). This measure was significantly correlated with the participant’s degree of attitude change from pretest, $r(163) = 0.29, p < 0.001$.

We then tested the effect of the manipulations on subjective persuasiveness by fitting linear regression models predicting subjective persuasiveness on the basis of medium, appeal and gender. There was not a significant three-way interaction between medium, appeal and gender, $F(4, 147) = 1.88, p = 0.12$. Furthermore, gender did not significantly interact with either medium or appeal individually, $F < 0.64, p > 0.53$. However, there was a significant interaction between medium and appeal, $F(4, 155) = 8.71, p < 0.001$. This interaction was driven by the fact that the text pathos condition was rated as substantially less persuasive than all the other conditions. Specifically, in the text condition, the pathos appeal was rated as significantly less persuasive than the ethos appeal, $b = 2.44, SE = 0.35, p < 0.001$, or the logos appeal, $b = 2.07, SE = 0.40, p < 0.001$. The pathos appeal was not rated worse than the other appeals in the video condition or the live condition, $b < 0.56, SE > 0.33, p > 0.096$. In the model including this interaction effect, participant/interviewee gender did not significantly associate with subjective persuasiveness, $b = 0.32, SE = 0.19, p = 0.090$.

Awareness of the persuasion intent

As the inoculation theory suggests, a person’s weak belief may be strengthened if the individual is facing a persuasion attack and tries to defend their belief (McGuire, 1964). So, we speculated that if the subject realized that the interviewee was trying to persuade him/her, then it might be more challenging to persuade the subject. We probed this by measuring the subject’s agreement (on a 5-point Likert scale) with the following question: “I believe that the person I interviewed was trying to change my opinion” (Q9). We observed no statistically significant effects of medium, appeal or gender on the perception that the interviewee was trying to persuade the participants.

The average and standard deviation (std) are calculated for all the questions and the results are shown in [Tables 2 and 3](#).

Discussion

The findings of our experiment have several major implications. First, the observed trivial difference between female–female and male–male conditions is intriguing. Our experiment design is remarkably similar to [Guadagno and Cialdini \(2002\)](#)’s. Specifically, both studies had the same persuasion setting – the subject was told about the comprehensive exam idea and asked to interview another person, a research assistant, using a scripted persuasion appeal. In their study, women with female partners were more likely to be persuaded in face-to-face interaction than via e-mail, whereas the medium did not play a role in the persuasion process for male–male pairs. In our study, however, we observed no difference between the gender groups in terms of the effects of the communication medium. One explanation is that it is the synchronicity of the communication that affects the persuasion power in a female–female setting. While [Guadagno and Cialdini \(2002\)](#) had synchronous and asynchronous conditions (face-to-face vs. email), our study only had synchronous situations (face-to-face, video-based chat and text-based chat). This could be one reason the two studies had different results. Another explanation is that these two studies are almost two decades apart, during which the relationship between gender and technology may have changed. Notably, computer use is not perceived as a masculine domain to the same extent today, because it has become so ubiquitous for most young people.

	Text-Patho (average/std)	Video-Etho (average/std)	Video-Logo (average/std)	Video-Patho (average/std)
N	16	12	21	23
opinion on the idea pre-test	3/1.21	3.42/1.24	3.43/0.98	3.13/1.22
opinion on the idea post-test	4.56/1.31	4.92/1.56	5.52/1.36	6.26/1.45
persuasion check: numerical (post pre)	1.56/1.36	1.5/2.32	2.10/1.73	2.13/1.69
interviewee's opinion (Q1)	6.56/0.73	6.67/0.49	6.71/0.56	6.78/0.42
similarity measure (between Q3's response and the interviewee's script)	0.33/0.09	0.32/0.06	0.45/0.08	0.33/0.07
Get along (Q4)	4.69/1.3	5.75/1.14	5.86/1.01	6.04/0.07
Good friends (Q5)	4.5/0.97	4.92/1	4.95/0.92	5.04/1.33
likeable (Q6)	5.31/1.2	6.25/0.97	6/0.95	6.13/0.87
persuasive (Q7)	3.31/1.3	6/0.74	5.90/1.14	5.39/1.37
strong argument (Q8)	2.94/1.34	6.08/0.9	6.29/0.85	5.78/1.17
engaging style (Q9)	4.63/1.54	5.25/1.86	5.38/1.50	5.87/1.14
trying to persuade (Q10)	3.25/1.65	3.67/2.06	3.81/1.57	3/1.62
genuine opinion (Q11)	5.25/1.39	6.17/1.03	5.57/1.29	6.13/1.42
Liking	4.83/1.04	5.64/0.93	5.60/0.82	5.74/0.89
Persuasiveness	3.13/1.24	6.04/0.75	6.10/0.92	5.59/1.18
persuasion check: descriptive	1 reverse 1 no change	2 reverse 2 no change	1 reverse 2 no change	2 reverse 3 no change

Source(s): Table by authors

Table 2.
Mean and standard
deviation of each
question in different
conditions (a)

	Live-Etho (average/std)	Live-Logo (average/ std)	Live-Patho (average/ std)	Text-Etho (average/ std)	Text-Logo (average/ std)
N	18	17	20	24	14
opinion on the idea pre-test	3.22/1.26	3.06/1.39	3.60/0.94	3.42/1.02	3.21/1.42
opinion on the idea post-test	5/1.61	5.65/1.22	5.05/1.19	4.79/1.67	6/0.96
persuasion check: numerical (post pre)	1.78/2.10	2.59/1.58	1.45/1.19	1.38/2.04	2.79/1.72
interviewee's opinion (Q1)	6.61/0.98	6.76/0.44	6.6/0.68	6.5/1.06	6.64/0.84
similarity measure (between Q3's response and the interviewee's script)	0.33/0.07	0.48/0.1	0.31/0.07	0.29/0.11	0.46/0.11
Get along (Q4)	5.83/0.86	6/0.94	5.75/0.97	4.79/1.53	5.14/1.23
Good friends (Q5)	4.67/1.14	4.82/1.19	4.8/1.15	3.96/1.68	4.14/1.23
likeable (Q6)	6.06/1.06	6.24/0.75	5.85/0.99	5.13/1.23	5.07/1.27
persuasive (Q7)	5.89/1.28	5.76/0.66	5.4/1.19	5.58/1.21	6.21/0.89
strong argument (Q8)	5.33/1.78	6.12/0.7	5.3/1.49	5.54/1.47	6.21/0.89
engaging style (Q9)	5.39/1.61	5.47/1.12	5.45/1.47	4.25/1.57	4.64/1.55
trying to persuade (Q10)	3.06/1.30	3.59/1.70	3.3/1.78	2.63/1.74	2.64/1.55
genuine opinion (Q11)	5.78/1.00	6.41/0.80	5.60/1.64	5.79/1.89	6.14/0.77
Liking	5.52/0.87	5.69/0.79	5.47/0.91	4.63/1.34	4.79/1.09
persuasiveness	5.61/1.35	5.94/0.50	5.35/1.31	5.56/1.28	6.21/0.83
persuasion check: descriptive	2: reverse direction; 3 no change	1: no change	5: no change	5: reverse; 2 no change	1: no change

Source(s): Table by authors

Table 3.
Mean and standard
deviation of each
question in different
conditions (b)

The second implication is related to research on combating the spread of disinformation on social media. With the prevalence of social media tools, widespread social media messaging and a considerable number of active users, the threat of the spread of unsubstantiated misinformation and disinformation is already having a significant negative impact on society (Carlson, 2020). Various efforts are being made to fight the spread of online misinformation and disinformation, such as developing fact-checking tools to verify the information (Ciampaglia *et al.*, 2015; Vlachos and Riedel, 2014).

Disinformation attempts to persuade people with distorted, misleading or fake information or data. Therefore, disinformation includes a form of logos appeal. In our logos appeal, we fabricated evidence – “an article . . . about job prospects for psych majors, and there was a survey showing that grades do not really reflect what graduating psych students know”. None of the participants tried to check the validity of this piece of information by asking a follow-up question for the source during the chat, presumably because the interview instructions encouraged them to follow the script exactly. However, during the debriefing, none of the 13 participants assigned to the logos appeal condition, who sensed that the comprehensive exam idea was fake, questioned this piece of evidence, either. Our analysis of the participants’ responses regarding why they felt the comprehensive exam idea was fake suggests that people used social cues and their prior beliefs when questioning the validity of the appeal, instead of doubting the truthfulness of the supporting evidence. Specifically, of the eleven participants who provided reasons as to why they felt the comprehensive exam idea was fake, six commented that the interview setup itself was “weird”. Two invalidated the exam idea because it contradicted their views: school policy shouldn’t change like this, and most majors have capstones, so there is no point to have the exam. Two thought the response “I do party a lot and have changed boyfriends/girlfriends twice already since I started college” was weird as this kind of response was not expected in such an interview, and one felt the interviewee’s response was too detailed and structured to be real. On the other hand, our study shows that logos ($M = 5.58$, $SD = 1.30$) was the most persuasive appeal of the three (compared to pathos, $M = 4.84$, $SD = 1.37$, and ethos, $M = 4.85$, $SD = 1.58$) independent of the communication media, $t > 2.6$, $p < 0.008$.

Furthermore, our similarity measures between the interview script and the participants’ open-ended reasoning of their views suggest that the influence of logo appeals can be deeper than changing people’s opinions they can also influence how people explain their changed view. These findings suggest it is helpful to use fact-checking tools to combat disinformation in cases where users may not have sufficient domain knowledge or may not realize the need to identify or examine the given information. Additionally, it may require more effort to negate the impact of the disinformation spread than correcting the information, as some users may not only believe false information but also may start to reason in ways like those presented in the disinformation messages.

We also acknowledge that the chosen interview topic may have an influence on the effectiveness of the logos appeal. The topic concerns a concrete practical matter – the consideration of a comprehensive exam in the Psychology department – and so lends itself to rational argument. An alternative topic that carries more emotion, e.g., persuading participants to select the most trustworthy person from three candidates (as the similar scenario in (Ma *et al.*, 2017)), may show different results. Would the logos appeal still be the most effective one? Future investigation is needed to explore the relationship between the type of topics and the effectiveness of different persuasion appeals.

The third implication from our study relates to the effectiveness of the pathos strategy – persuading people by manipulating their emotions. In our study, participants’ initial position was slightly opposed to the exam idea, on average ($M = 3.29$, $SD = 1.16$), but their position after the persuasion attempt favored the idea ($M = 5.18$, $SD = 1.43$). Several participants experienced a change in their views in our experiment. Hullet’s (2005) meta-analysis on the

relationship between mood and an argument's persuasive power suggests that while stimulating positive emotion is more effective for establishing or reinforcing views, it is negative emotion stimulation that is more persuasive for inducing view changes. This analysis could explain the observed ineffectiveness of the pathos strategy in our study, as it aimed to arouse positive emotions towards the position of in favor of the comprehensive exam. Another supporting explanation comes from studies on online discussions. Specifically, these studies suggest that in text-based online discussions on social media, persuasive messages are more likely to have a negative sentiment than nonpersuasive ones, regardless of the behavioral outcome (i.e., view change, view establishment or view reinforcement) (Xiao, 2018).

Our experiment further reveals that interviewees were perceived as the least persuasive when employing the pathos strategy in the text-based chat. Initially, the dual-process models of persuasion (Chaiken, 1987; Petty and Cacioppo, 1986) can help explain this finding. According to these models, a persuasion attempt may succeed through two paths: individuals carefully consider the information, including the quality of the argument or rely on heuristic cues to make judgments (Goldstein *et al.*, 2008). The manipulation of emotions facilitates heuristic processing which is not as profound as systematic processing. In the text-based chat condition, the indirect nature of communication may slow down heuristic processing and/or encourage systematic processing compared to face-to-face and video-based chat conditions. People have more time to carefully formulate their thoughts or expressions in the text-based condition.

A caveat to this explanation is our observation that medium did not make a difference in measures such as changing the subject's opinion and influencing the subject's reasoning or justification for their opinion. Interestingly, our findings indicate that interviewees in the text condition were the least liked by the subjects. It is possible that this lack of liking in the text-based condition contributed to the reported low receptivity to the emotion appeal (i.e., the pathos persuasion act). Liking is an influential factor in the persuasion process (Cialdini, 2007) and is related to the subject's emotional state during the experiment. We speculate that the positive emotion cues in the pathos script may have come across less positively in the text condition, either due to the absence of nonverbal cues or because the interviewee was less likeable. Additionally, the discrepancy could be attributed to esoteric features of the script, given that we only have one example, lacking evidence for generalizability.

Moreover, the observation that medium played a limited role in the persuasion process in our experiment is intriguing. Video and face-to-face settings, allowing for more nonverbal communications, were expected to provide additional signals to enhance persuasion power. For instance, studies have shown that people exhibit different gaze patterns when making persuasive versus nonpersuasive statements (Vrzakova *et al.*, 2014). In their experiment comparing voice- and text-based virtual assistant interactions, Ischen *et al.* (2022) found that voice communication can increase persuasion by being more cognitively demanding. Jung *et al.* (2022)'s study of clubhouse, a voice-based social media platform, suggested that voice is effective for establishing social relationships, introducing more interactivity and intimacy, which according to Cialdini's influence principles, can enhance persuasion power compared to text-based interactions. However, the medium did not affect the persuasion outcome in our experiment. This aligns with Bradner and Mark's (2002) finding that video and instant messaging is indistinguishable in terms of persuasion outcome in a one-to-one shared decision-making scenario. Applying the ELM, we speculate that in both studies, participants focused more on the presented information than contextual cues. Therefore, the additional signals through facial expression, gesture or voice may not have been as influential in shaping their attitudes. Consequently, we did not observe the medium's effect in the studies. This presumed emphasis on information may also help explain the superior performance of logos in our persuasion experiment. Interestingly, online environments allow the use of

multimedia and communication formats beyond plain text in the communication process. [Xiao et al.'s \(2019\)](#) experiment demonstrates that the use of an abstract comic form has more persuasion power than plain text in charitable donation situations. A future direction from this study is to examine whether and how various formats of logos, pathos and ethos affect the online persuasion process and outcome.

One significant limitation of our study is that our subjects are college students, and therefore, are not representative of Internet users in general. Future research should explore the effectiveness of these three types of persuasion strategies on people from diverse backgrounds. For instance, while logos proved to be the most effective strategy in persuading college students in our study, pathos or ethos might be more effective when attempting to persuade people with diverse backgrounds. Despite meeting the requirements for statistical testing, our sample size remains relatively small due to constraints on time, personnel and funding.

We also acknowledge the conceptual and empirical challenges of comparing the effectiveness of the three persuasion strategies separately. While our logos appeal focuses on presenting logical thinking and evidence to support the claim (e.g., “an article . . . about job prospects for psych majors”; please refer to [Appendix](#) for script details), it remains unknown whether this persuasion strategy also triggers certain emotions or memories in the subject (e.g., the subject may have read a similar article before), leading to persuasion under the influence of emotion or belief in the credibility of the script rather than logical persuasion. Additionally, in general applications of the pathos strategy, various emotions may be aroused or manipulated to persuade people. In our study, the aim of the pathos script was solely to evoke the subject's positive feeling about the idea of requiring students to take a comprehensive exam. The observation that the pathos script was the least persuasive in this experiment cannot be generalized to all pathos strategy situations, particularly those involving quite different emotional content.

Conclusion

Human social interaction involves a near-constant exchange and updating of information and ideas. With the widespread growth of online communities, a communication revolution is underway, necessitating the updating of traditional persuasion strategies to suit modern media. While the distinction between appeals to logic, emotion and authority is ancient, ideas developed for live face-to-face conversation processes must be tested in new media. Past studies have analyzed online communication content to understand the use of persuasion strategies. The current findings suggest that participants have a preference among the three persuasion strategies: logos, pathos and ethos.

Using a 3×3 between-subject factorial design, we tested these three strategies in one-to-one chats (female–female or male–male pairs) through three communication media: face-to-face interactions, computer-mediated video or real-time computer-mediated text chat. Participants were recruited from the subject pool at the Department of Psychology of a major university in New York state, USA. The experiment focused on a personally relevant topic in the experiment – the idea of requiring Psychology students to take a comprehensive exam to graduate. Our results indicate a trivial difference between female–female and male–male pairs in terms of persuasion outcomes. Across media types, the logos appeal proved most effective in persuading participants. Additionally, the explanations provided by participants were most similar to the persuasion scripts in this condition compared to the other two appeal conditions. Participants responded particularly negatively to the pathos appeal in the text-based chat condition, although this did not manifest in substantially different attitude change outcomes.

In the previous section, we offer our explanations of these observations, discuss the implications and call for future investigations. We acknowledge a major limitation of our study, wherein our subjects are college students thus are not representative of Internet users

in general. Furthermore, we tested only one content area, which is not representative of persuasive appeals in general. Future research should explore these three types of persuasion strategies on people from diverse backgrounds communicating about diverse topics.

Appeals to
logic, authority
and emotion

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Appendix

Logo condition

I think it would be great if this became a requirement for the degree. I was reading an article the other day about job prospects for psych majors, and there was a survey showing that grades do not really reflect what graduating psych students know. Some students know a lot of the important details about research methods and psychological concepts and some don't but employers cannot tell the difference because the transcripts do not necessarily show. I am not surprised by this because even for someone like me, who studies hard and always gets an A or A-in my psych classes, I tend to forget what I have learned from previous semesters. I think this comprehensive exam would provide a good way to signal the quality of psych graduates from our university. It will force us to review the course materials from past semesters in our senior year. Then, when we pass the exam, we are proving that we have a shared level of

knowledge. Plus, preparing for the comprehensive exam will help students prepare for job interviews at the same time, and it will give us something distinctive to talk about when we're explaining why we're qualified. So, I think it makes sense to implement this idea and we should do so.

Appeals to
logic, authority
and emotion

Patho condition

(research assistant brings in excitement feeling to the interview, in text, use emoticon)

I like this idea! First, it will make my mom so proud of me. Hahaha. Seriously, my mom thinks strangely how college life is like. She is quite worried that I am not learning enough of the important subject matter. I mean I do party a lot and have changed boyfriends [if the research assistant is female]/girlfriends [if the research assistant is male] twice already since I started college. But I also study hard and get good grades in courses. But she is still worried. If we need to pass a comprehensive exam in order to get our degree, I am sure that I will have to study a lot in my senior year, but in the end, my mom will be so impressed and so proud of me when I pass it at the end of the program. Making our parents proud of us is our responsibility. Don't you think? (Smile) Also, I love taking challenges because it feels so rewarding after I've done with it. Imagine, a comprehensive exam that tests all major things that you've learned in the last four years. Wouldn't that be a great accomplishment yourself? We can also say to those who think Psychology is not a serious major: look, there is a ton of things to learn here and we have a serious comprehensive exam that shows it (smile).

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Etho condition

I think it's a good idea and we should do it. I recently transferred from a school where we had comprehensive exams. Our undergraduate advisor there had 15 years of experience coaching students through them. The first meeting we had covered all kinds of school policies, degree requirements, etc. She explained this comprehensive exam requirement to me and said that they are a valuable asset for the students. I had some friends there who at that time were graduating, and they all passed the exam. They said that it was a very beneficial experience because reviewing the course materials and passing the exam helped them in job seeking and job interviews. At SU, I started my minor last semester at School of Information Study. The school is planning the same thing. Basically, there are eight required courses in this minor, five for Technology and three for Management. The school plans to offer a test to students in their last year to test their knowledge related to these courses, like information technology (IT) Management and Administration, Database Management and Information Analysis. While the school has not decided to do it, my academic adviser at the school said they have collected a lot of feedback from current juniors and recent graduates and majority of them supported this idea.

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